

FRP FNPT Couplings

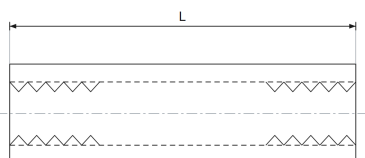


Female NPT threaded couplings, threaded on both ends, moulded in Derakane 411-350 or Derakane 470-300. Supplied in three standard lengths per size.

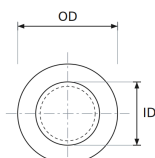
½ – 6 in. SIZE RANGE

4 · 6 · 8 in. STANDARD LENGTHS

FNPT BOTH ENDS, ASME B1.20.1



FNPT both ends



ID nominal inside diameter
L overall length

OD outside diameter
FNPT female NPT taper thread

ID nominal inside dia.	OD outside diameter	L available lengths	thread both ends
in	in	in	—
1/2	1-1/4	4 · 6 · 8	FNPT
3/4	1-7/16	4 · 6 · 8	FNPT
1	1-3/4	4 · 6 · 8	FNPT
1-1/4	2-1/16	4 · 6 · 8	FNPT
1-1/2	2-5/16	4 · 6 · 8	FNPT
2	2-7/8	4 · 6 · 8	FNPT
2-1/2	3-3/8	4 · 6 · 8	FNPT
3	4	4 · 6 · 8	FNPT
4	5	4 · 6 · 8	FNPT
6	7-3/8	4 · 6 · 8	FNPT

NOTES

- Female NPT taper pipe thread to ASME B1.20.1 on both ends.
- Pressure capability depends on size, length and operating conditions and must be confirmed for the selected configuration — contact FRP Depots before specifying.
- Laminate construction and corrosion barrier as described on page 03.
- Thread sealant must be compatible with the service chemistry. PTFE tape suits most services.

Not every size is held in stock at all times. Confirm availability and lead time with sales@frpdepots.com.

STANDARD PRODUCT RANGE AT A GLANCE

PRODUCT component family	SIZE RANGE nominal inside diameter	PRESSURE CLASSES as tabulated	CONSTRUCTION method	RESIN stocked	PAGE see
—	in	psi	—	—	—
Stub flanges	½ – 36	50 · 75 · 150	Contact moulded	D411 / D470	04
Manways & covers	20 – 36	0.5 · 15 · 25 · 75	Contact moulded	D411 / D470	05
90° elbows	1 – 24	75 · 150	Contact moulded	D411 / D470	06
Filament-wound pipe	1 – 36	150	Filament wound	D411 / D470	07
FNPT couplings	½ – 6	consult factory	Moulded	D411 / D470	08

Stub flanges are not stocked in NPS 1-1/4 or NPS 5. Pipe is not stocked in NPS 2-1/2. Not every size is offered in every pressure class — see the section tables. Sizes and pressure classes outside these ranges are available as special order.

SYSTEM PRESSURE RATING

Where a system is assembled from FRP Depots components, the system rating is governed by the lowest-rated component. Pipe is rated 150 psi in all sizes, while stub flanges above NPS 18 and elbows above NPS 16 are rated 50–75 psi. Systems above NPS 16 must be de-rated accordingly, or the higher-rated component specified as a special order.

BEYOND THE STANDARD RANGE

Made to your drawing

Non-standard diameters and pressure classes, special drilling patterns, non-standard stub lengths, 45° elbows, tees and reducers, and flanged, bell-and-spigot or butt-and-strap joint preparations are all quoted on request.

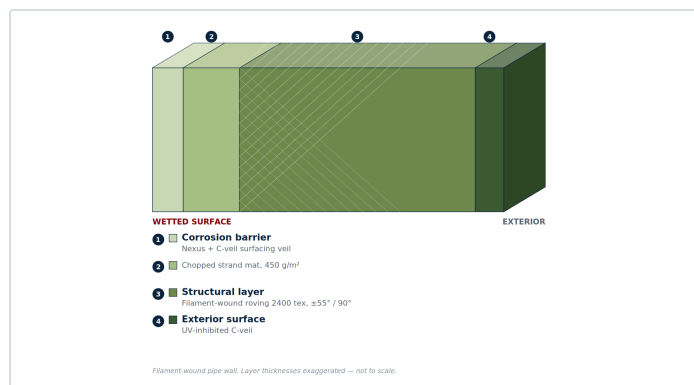
Tell us the service, not just the size

Send the chemical, concentration, temperature and operating pressure with your enquiry. Resin selection and corrosion-barrier thickness are driven by the service, and getting them right at quotation stage is far cheaper than getting them wrong in the field.

Resins, Laminate & Standards

Every component is built as a layered laminate. The layer names below are those used by ASTM C582 and ASME RTP-1 and are the common vocabulary of the industry. What differs between products is not the structure but the ply build-up and the way the structural layer is laid down. Wall thicknesses quoted throughout this catalogue are total thicknesses and include the corrosion barrier.

LAMINATE CONSTRUCTION



Filament-wound pipe wall. Layer thicknesses not to scale.

Inner surface

A resin-rich layer reinforced with a surfacing veil — glass, synthetic or carbon. Nominally 0.010–0.020 in. This is the surface the process fluid touches.

Interior layer

Chopped strand mat, two plies minimum. With the inner surface it forms the **corrosion-resistant barrier** — the part of the wall that does the chemical work, and the part excluded from the strength calculation.

Structural layer

Carries the pressure and the mechanical load. Filament wound on pipe; contact moulded — hand lay-up — on every fitting. The first ply against the barrier is always mat.

Outer surface

A resin coat, paraffinated for a full surface cure. UV-inhibited exterior standard on pipe, available on fittings.

Ply-by-ply build-up differs by product and method. The filament-wound pipe schedule is tabulated on page 07; fitting schedules are held by the manufacturer — see the scope note below.

STANDARDS OF MANUFACTURE AND REFERENCE

CGSB 41-GP-22	Process equipment — reinforced polyester, glass fibre. Governing Canadian fabrication standard; covers tanks, pipe and ductwork, and establishes the 10:1 safety factor.
ASTM D5421	Contact-Molded "Fiberglass" Flanges. Governs the stub flanges and manway flanges.
ASME RTP-1	Referenced for bolt-hole drilling compatibility with RTP-1 vessel nozzle and manway patterns only — see the scope note below.
ASME B16.1 Cl. 125	Bolt-hole drilling, NPS 30 and 36.
ASME B18.22.1	Type A Narrow plain washers — governs back-face (BF) diameters.

RESIN SYSTEMS

Derakane 411-350

Bisphenol-A epoxy vinyl ester

HDT	105 °C / 220 °F	TENSILE	86 MPa / 12,000 psi
MODULUS	3,200 MPa	ELONGATION	5–6 %

The standard offering. Broad resistance to acids, alkalis, bleaches and solvents.

Derakane 470-300

Epoxy novolac vinyl ester

HDT	150 °C / 300 °F	TENSILE	85 MPa / 12,500 psi
MODULUS	3,600 MPa	ELONGATION	3–4 %

Specified where elevated temperature, solvent or oxidiser service (chlorine, flue gas) demands higher thermal and chemical resistance.

HDT is a material property, not a service rating. Maximum service temperature is governed by the chemical environment — confirm against the resin manufacturer's chemical resistance guide before specifying.

DESIGN BASIS

- Minimum **10:1 safety factor** on the structural laminate, per CGSB 41-GP-22.
- The corrosion-resistant barrier is excluded from the strength calculation — the safety factor applies to the structural wall only.
- Filament-wound structural layers are designed on a **2:1 hoop-to-axial** stress ratio.

GASKETS & BOLTING

- 1/8 in. full-face elastomeric gasket, Shore A60 ± 5, minimum design seating stress (y) 100 psi, gasket factor (m) 1.
- Flat-face mating flanges only — never bolt FRP to a raised-face metallic flange without a full-face adaptor.
- Tighten in a star pattern in equal stages. FRP flanges are damaged by over-torquing — ask us for torque values.

STANDARDS SCOPE

- ASME RTP-1 is a vessel standard** rated to 15 psig. It does not cover piping or pipe fittings. Where we cite it, we mean bolt-hole drilling compatibility with RTP-1 vessel nozzle and manway patterns — nothing here is offered as an RTP-1 certified vessel component.
- We are a stockist. Item-specific laminate schedules and conformance documentation come from the manufacturer against a specific order — ask at enquiry stage.



FRP DEPOTS

CANADA'S DIRECT SOURCE FOR FRP COMPONENTS

How to order

Send your enquiry to sales@frpdepots.com or call **613-499-4950**. Include the following and we will quote from stock where we can.

- | | |
|---|---|
| 01 Product type and nominal diameter | 02 Pressure rating required |
| 03 Resin system — D411 or D470 | 04 Quantity and length (pipe, couplings) |
| 05 Any non-standard drilling, stub length or joint preparation | 06 Drawings or job specification, if you have them |

Full dimensional data, tolerances and design basis are published in **FRPD-SPEC-001 Revision R3**. Request the current revision with your enquiry.

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